# Bayesian Optimization Report: Function 2

## GP Health Summary per Iteration

| Iteration | GP Health | Status |
|---|---|---|
| 1 | 0.746 | Good |
| 2 | 0.746 | Good |
| 3 | 0.746 | Good |
| 4 | 0.746 | Good |
| 5 | 0.745 | Good |
| 6 | 0.745 | Good |
| 7 | 0.744 | Good |
| 8 | 0.744 | Good |
| 9 | 0.744 | Good |
| 10 | 0.743 | Good |
| 11 | 0.743 | Good |
| 12 | 0.713 | Good |
| 13 | 0.713 | Good |
| 14 | 0.742 | Good |
| 15 | 0.741 | Good |
| 16 | 0.740 | Good |
| 17 | 0.709 | Good |
| 18 | 0.739 | Good |
| 19 | 0.707 | Good |
| 20 | 0.706 | Good |
| 21 | 0.738 | Good |
| 22 | 0.704 | Good |
| 23 | 0.703 | Good |
| 24 | 0.702 | Good |
| 25 | 0.702 | Good |
| 26 | 0.700 | Good |
| 27 | 0.700 | Medium |
| 28 | 0.699 | Medium |
| 29 | 0.698 | Medium |
| 30 | 0.697 | Medium |

## GP Sigma Summary

| Iteration | Mean Sigma | Min Sigma | Max Sigma |
|---|---|---|---|
| 1 | 0.2565 | 0.2536 | 0.2570 |
| 2 | 0.2494 | 0.2459 | 0.2502 |
| 3 | 0.2420 | 0.2381 | 0.2426 |
| 4 | 0.2351 | 0.2295 | 0.2363 |
| 5 | 0.2296 | 0.2242 | 0.2309 |
| 6 | 0.2248 | 0.2190 | 0.2265 |
| 7 | 0.2187 | 0.2109 | 0.2210 |
| 8 | 0.2138 | 0.2071 | 0.2161 |
| 9 | 0.2094 | 0.2031 | 0.2115 |
| 10 | 0.2041 | 0.1951 | 0.2074 |
| 11 | 0.2018 | 0.1986 | 0.2032 |
| 12 | 0.1866 | 0.1713 | 0.1986 |
| 13 | 0.1823 | 0.1705 | 0.1928 |
| 14 | 0.1875 | 0.1781 | 0.1925 |
| 15 | 0.1826 | 0.1692 | 0.1892 |
| 16 | 0.1793 | 0.1670 | 0.1870 |
| 17 | 0.1670 | 0.1516 | 0.1808 |
| 18 | 0.1736 | 0.1611 | 0.1816 |
| 19 | 0.1597 | 0.1462 | 0.1753 |
| 20 | 0.1559 | 0.1421 | 0.1726 |
| 21 | 0.1697 | 0.1648 | 0.1741 |
| 22 | 0.1517 | 0.1382 | 0.1699 |
| 23 | 0.1482 | 0.1334 | 0.1649 |
| 24 | 0.1434 | 0.1257 | 0.1632 |
| 25 | 0.1401 | 0.1231 | 0.1604 |
| 26 | 0.1396 | 0.1242 | 0.1570 |
| 27 | 0.1342 | 0.1146 | 0.1563 |
| 28 | 0.1337 | 0.1166 | 0.1553 |
| 29 | 0.1308 | 0.1099 | 0.1522 |
| 30 | 0.1294 | 0.1113 | 0.1514 |

## Best Results Per Iteration

| Iteration | Best Input | Best Output | Kernel | Acquisition | GP Health | Anomalies |
|---|---|---|---|---|---|---|
| 1 | [0.80285345 0.00463202] | 0.3404539148105217 | Matern(ν=0.5) | UCB | 0.746 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 2 | [0.58934875 0.97920813] | 0.3385456456248681 | Matern(ν=0.5) | UCB | 0.746 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 3 | [0.72225607 0.36159478] | 0.29984402390717757 | Matern(ν=0.5) | UCB | 0.746 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 4 | [0.32247376 0.93814458] | 0.2893011556466782 | Matern(ν=0.5) | UCB | 0.746 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 5 | [0.81959581 0.14112771] | 0.34981274244562116 | Matern(ν=0.5) | UCB | 0.745 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 6 | [0.7331814 0.80374818] | 0.33001890539315304 | Matern(ν=0.5) | UCB | 0.745 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 7 | [0.07046804 0.09621932] | 0.2912062042230842 | Matern(ν=0.5) | UCB | 0.744 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 8 | [0.71017944 0.52349735] | 0.2927652282394036 | Matern(ν=0.5) | UCB | 0.744 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 9 | [0.14190553 0.66245697] | 0.2935050080190014 | Matern(ν=0.5) | UCB | 0.744 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 10 | [0.34667299 0.51895629] | 0.29170945537067994 | Matern(ν=0.5) | UCB | 0.743 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |

| | | | | | | |
|---|---|---|---|---|---|---|
| 11 | [0.58779188 0.20614146] | 0.3131161161650342 | Matern(ν=0.5) | UCB | 0.743 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 12 | [0.97877216 0.47199465] | 0.22520700357215817 | Matern(ν=0.5) | UCB | 0.713 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 13 | [0.99983039 0.7013682 ] | 0.2528521366891335 | Matern(ν=0.5) | UCB | 0.713 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 14 | [0.8575571 0.27789703] | 0.29938882858122484 | Matern(ν=0.5) | UCB | 0.742 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 15 | [0.78767644 0.89986963] | 0.36959116411118204 | Matern(ν=0.5) | UCB | 0.741 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 16 | [0.00110087 0.74887304] | 0.2918143888561307 | Matern(ν=0.5) | UCB | 0.740 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 17 | [0.2624169 0.75586084] | 0.24528515398367465 | Matern(ν=0.5) | UCB | 0.709 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 18 | [0.54781091 0.468463 ] | 0.2894077343589779 | Matern(ν=0.5) | UCB | 0.739 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 19 | [0.6551747 0.00231796] | 0.35088503052668685 | Matern(ν=0.5) | UCB | 0.707 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 20 | [0.44923556 0.99942523] | 0.24765090473696783 | Matern(ν=0.5) | UCB | 0.706 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |

| | | | | | | |
|---|---|---|---|---|---|---|
| 21 | [0.79009905 0.9918783 ] | 0.32152138606066316 | Matern(ν=0.5) | UCB | 0.738 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 22 | [0.01416684 0.57006126] | 0.23230727683938995 | Matern(ν=0.5) | UCB | 0.704 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 23 | [0.63702384 0.2798 ] | 0.31935907777666445 | Matern(ν=0.5) | UCB | 0.703 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 24 | [0.27654873 0.62544252] | 0.2597044767667247 | Matern(ν=0.5) | UCB | 0.702 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 25 | [0.60069806 0.90029134] | 0.33247426975112976 | Matern(ν=0.5) | UCB | 0.702 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 26 | [0.83792384 0.49356438] | 0.24982272028511943 | Matern(ν=0.5) | UCB | 0.700 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 27 | [0.00203878 0.02881737] | 0.2577658353121368 | Matern(ν=0.5) | UCB | 0.700 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 28 | [0.97585766 0.79781692] | 0.27922203957344344 | Matern(ν=0.5) | UCB | 0.699 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 29 | [0.10922292 0.80114391] | 0.24501329634285912 | Matern(ν=0.5) | UCB | 0.698 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |
| 30 | [0.45383662 0.50255604] | 0.2744265527124815 | Matern(ν=0.5) | UCB | 0.697 | {'type': 'lengthscale_collapse', 'severity': 0.9, 'details': {'lengthscales': [1e-05]}} |

## Acquisition Values Per Iteration

### *Iteration 1*

| Method | Max Value |
|---|---|
| UCB | 0.842218 |
| EI | 0.004561 |
| PI | 0.042109 |
| THOMPSON | 0.675445 |
| KG | 0.004561 |
| ENTROPY | 1.468415 |
| PORTFOLIO | 1.029832 |

### *Iteration 2*

| Method | Max Value |
|---|---|
| UCB | 0.474184 |
| EI | 0.003880 |
| PI | 0.037593 |
| THOMPSON | 0.781371 |
| KG | 0.003880 |
| ENTROPY | 1.391261 |
| PORTFOLIO | 0.924484 |

### *Iteration 3*

| Method | Max Value |
|---|---|
| UCB | 0.448214 |
| EI | 0.002574 |
| PI | 0.026960 |
| THOMPSON | 0.677599 |
| KG | 0.002574 |
| ENTROPY | 1.404164 |
| PORTFOLIO | 0.919532 |

### *Iteration 4*

| Method | Max Value |
|---|---|
| UCB | 0.445565 |
| EI | 0.002255 |
| PI | 0.024818 |

| | |
|---|---|
| THOMPSON | 0.657448 |
| KG | 0.002255 |
| ENTROPY | 1.394331 |
| PORTFOLIO | 0.818907 |

### *Iteration 5*

| Method | Max Value |
|---|---|
| UCB | 0.462916 |
| EI | 0.002740 |
| PI | 0.030082 |
| THOMPSON | 0.592889 |
| KG | 0.002740 |
| ENTROPY | 1.306278 |
| PORTFOLIO | 0.768957 |

### *Iteration 6*

| Method | Max Value |
|---|---|
| UCB | 0.444830 |
| EI | 0.002014 |
| PI | 0.023291 |
| THOMPSON | 0.769871 |
| KG | 0.002014 |
| ENTROPY | 1.331980 |
| PORTFOLIO | 0.738489 |

### *Iteration 7*

| Method | Max Value |
|---|---|
| UCB | 0.431679 |
| EI | 0.001523 |
| PI | 0.018627 |
| THOMPSON | 0.771142 |
| KG | 0.001523 |
| ENTROPY | 1.267731 |
| PORTFOLIO | 0.711966 |

### *Iteration 8*

| Method | Max Value |
|---|---|
| UCB | 0.422443 |
| EI | 0.001247 |
| PI | 0.015822 |
| THOMPSON | 0.666186 |
| KG | 0.001247 |
| ENTROPY | 1.260120 |
| PORTFOLIO | 0.659984 |

### *Iteration 9*

| Method | Max Value |
|---|---|
| UCB | 0.422886 |
| EI | 0.001151 |
| PI | 0.015103 |
| THOMPSON | 0.642324 |
| KG | 0.001151 |
| ENTROPY | 1.353627 |
| PORTFOLIO | 0.614695 |

### *Iteration 10*

| Method | Max Value |
|---|---|
| UCB | 0.423083 |
| EI | 0.000990 |
| PI | 0.013503 |
| THOMPSON | 0.675509 |
| KG | 0.000990 |
| ENTROPY | 1.341745 |
| PORTFOLIO | 0.605801 |

### *Iteration 11*

| Method | Max Value |
|---|---|
| UCB | 0.413335 |
| EI | 0.000873 |
| PI | 0.012112 |

| | |
|---|---|
| THOMPSON | 0.615430 |
| KG | 0.000873 |
| ENTROPY | 1.273434 |
| PORTFOLIO | 0.553784 |

### *Iteration 12*

| Method | Max Value |
|---|---|
| UCB | 0.364333 |
| EI | 0.000261 |
| PI | 0.004294 |
| THOMPSON | 0.537656 |
| KG | 0.000261 |
| ENTROPY | 1.100490 |
| PORTFOLIO | 0.519214 |

### *Iteration 13*

| Method | Max Value |
|---|---|
| UCB | 0.359730 |
| EI | 0.000216 |
| PI | 0.003697 |
| THOMPSON | 0.527571 |
| KG | 0.000216 |
| ENTROPY | 1.089827 |
| PORTFOLIO | 0.459028 |

### *Iteration 14*

| Method | Max Value |
|---|---|
| UCB | 0.401548 |
| EI | 0.000516 |
| PI | 0.008120 |
| THOMPSON | 0.670568 |
| KG | 0.000516 |
| ENTROPY | 1.246131 |
| PORTFOLIO | 0.492506 |

### *Iteration 15*

| Method | Max Value |
|---|---|
| UCB | 0.427015 |
| EI | 0.000660 |
| PI | 0.010872 |
| THOMPSON | 0.552633 |
| KG | 0.000660 |
| ENTROPY | 1.127086 |
| PORTFOLIO | 0.433898 |

### *Iteration 16*

| Method | Max Value |
|---|---|
| UCB | 0.383005 |
| EI | 0.000356 |
| PI | 0.005768 |
| THOMPSON | 0.576871 |
| KG | 0.000356 |
| ENTROPY | 1.140042 |
| PORTFOLIO | 0.416444 |

### *Iteration 17*

| Method | Max Value |
|---|---|
| UCB | 0.343105 |
| EI | 0.000092 |
| PI | 0.001766 |
| THOMPSON | 0.456368 |
| KG | 0.000092 |
| ENTROPY | 0.999067 |
| PORTFOLIO | 0.384823 |

### *Iteration 18*

| Method | Max Value |
|---|---|
| UCB | 0.383219 |
| EI | 0.000264 |
| PI | 0.004524 |

| | |
|---|---|
| THOMPSON | 0.551824 |
| KG | 0.000264 |
| ENTROPY | 1.079394 |
| PORTFOLIO | 0.400600 |

### *Iteration 19*

| Method | Max Value |
|---|---|
| UCB | 0.373308 |
| EI | 0.000122 |
| PI | 0.002694 |
| THOMPSON | 0.409505 |
| KG | 0.000122 |
| ENTROPY | 0.973266 |
| PORTFOLIO | 0.334508 |

### *Iteration 20*

| Method | Max Value |
|---|---|
| UCB | 0.338294 |
| EI | 0.000047 |
| PI | 0.001070 |
| THOMPSON | 0.476170 |
| KG | 0.000047 |
| ENTROPY | 0.966362 |
| PORTFOLIO | 0.326732 |

### *Iteration 21*

| Method | Max Value |
|---|---|
| UCB | 0.385482 |
| EI | 0.000229 |
| PI | 0.004271 |
| THOMPSON | 0.516977 |
| KG | 0.000229 |
| ENTROPY | 1.074289 |
| PORTFOLIO | 0.365902 |

### *Iteration 22*

| Method | Max Value |
|---|---|
| UCB | 0.333785 |
| EI | 0.000036 |
| PI | 0.000787 |
| THOMPSON | 0.402984 |
| KG | 0.000036 |
| ENTROPY | 0.981957 |
| PORTFOLIO | 0.317961 |

### *Iteration 23*

| Method | Max Value |
|---|---|
| UCB | 0.339994 |
| EI | 0.000028 |
| PI | 0.000690 |
| THOMPSON | 0.435098 |
| KG | 0.000028 |
| ENTROPY | 0.969200 |
| PORTFOLIO | 0.331286 |

### *Iteration 24*

| Method | Max Value |
|---|---|
| UCB | 0.311006 |
| EI | 0.000016 |
| PI | 0.000390 |
| THOMPSON | 0.496326 |
| KG | 0.000016 |
| ENTROPY | 0.953906 |
| PORTFOLIO | 0.302257 |

### *Iteration 25*

| Method | Max Value |
|---|---|
| UCB | 0.339981 |
| EI | 0.000016 |
| PI | 0.000403 |

| | |
|---|---|
| THOMPSON | 0.397457 |
| KG | 0.000016 |
| ENTROPY | 0.939760 |
| PORTFOLIO | 0.286862 |

### *Iteration 26*

| Method | Max Value |
|---|---|
| UCB | 0.313942 |
| EI | 0.000012 |
| PI | 0.000311 |
| THOMPSON | 0.459108 |
| KG | 0.000012 |
| ENTROPY | 0.913476 |
| PORTFOLIO | 0.280043 |

### *Iteration 27*

| Method | Max Value |
|---|---|
| UCB | 0.316633 |
| EI | 0.000011 |
| PI | 0.000271 |
| THOMPSON | 0.441946 |
| KG | 0.000011 |
| ENTROPY | 0.893304 |
| PORTFOLIO | 0.269277 |

### *Iteration 28*

| Method | Max Value |
|---|---|
| UCB | 0.304318 |
| EI | 0.000007 |
| PI | 0.000183 |
| THOMPSON | 0.397526 |
| KG | 0.000007 |
| ENTROPY | 0.881408 |
| PORTFOLIO | 0.277090 |

### *Iteration 29*

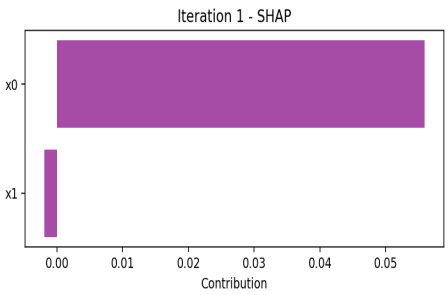
| Method | Max Value |
|---|---|
| UCB | 0.298968 |
| EI | 0.000008 |
| PI | 0.000199 |
| THOMPSON | 0.462951 |
| KG | 0.000008 |
| ENTROPY | 0.929118 |
| PORTFOLIO | 0.273533 |

### *Iteration 30*

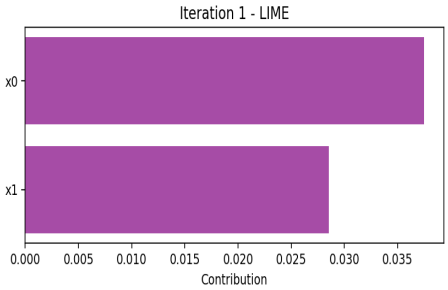
| Method | Max Value |
|---|---|
| UCB | 0.299507 |
| EI | 0.000006 |
| PI | 0.000163 |
| THOMPSON | 0.387485 |
| KG | 0.000006 |
| ENTROPY | 0.865167 |
| PORTFOLIO | 0.251505 |

## Feature Explanations (SHAP / LIME)

### Iteration 1 - SHAP




### Iteration 1 - LIME



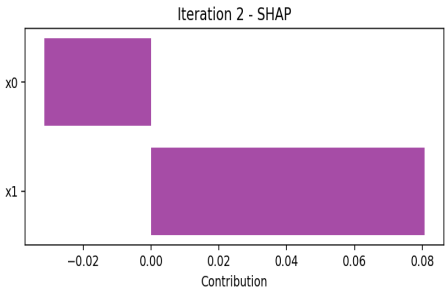

## Iteration 1 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | 0.05593 |
| x1 | -0.00191 |

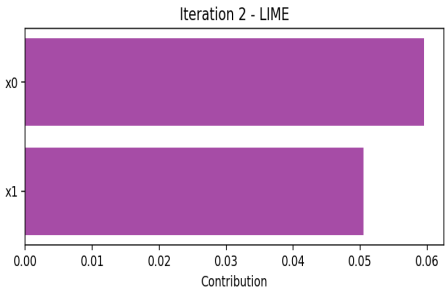
## Iteration 1 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.03748 |
| x1 | 0.02856 |

## Iteration 2 - SHAP




## Iteration 2 - LIME



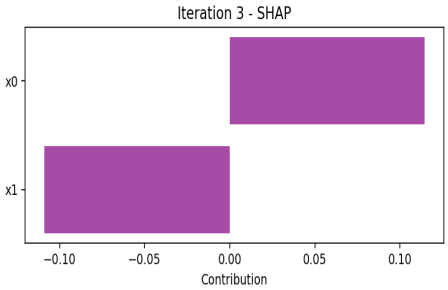

## Iteration 2 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.03147 |
| x1 | 0.08072 |

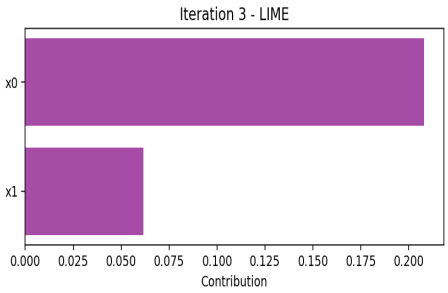
## Iteration 2 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.05952 |
| x1 | 0.05052 |

## Iteration 3 - SHAP




## Iteration 3 - LIME



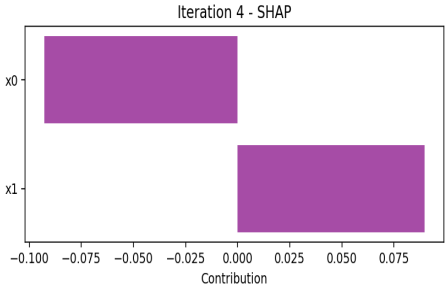

## Iteration 3 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | 0.11459 |
| x1 | -0.10891 |

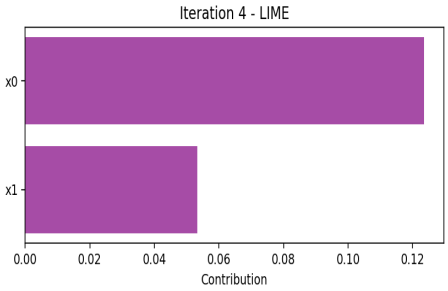
## Iteration 3 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.20782 |
| x1 | 0.06151 |

## Iteration 4 - SHAP




## Iteration 4 - LIME



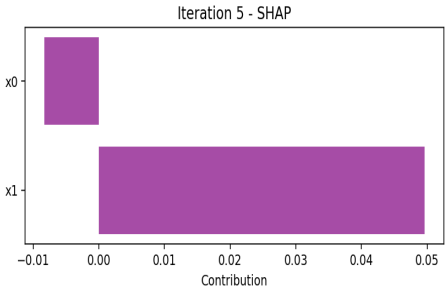

## Iteration 4 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.09274 |
| x1 | 0.08981 |

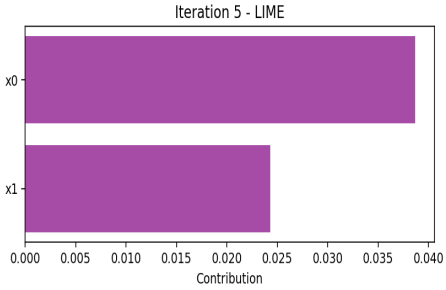
## Iteration 4 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.12346 |
| x1 | 0.05321 |

## Iteration 5 - SHAP



## Iteration 5 - LIME



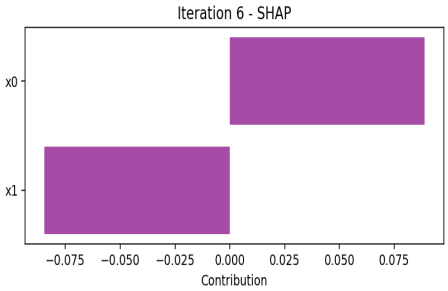

## Iteration 5 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.00831 |
| x1 | 0.04964 |

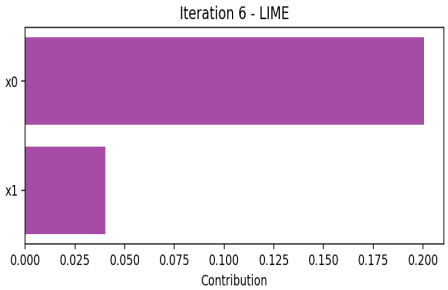
## Iteration 5 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.03868 |
| x1 | 0.02430 |

## Iteration 6 - SHAP




## Iteration 6 - LIME



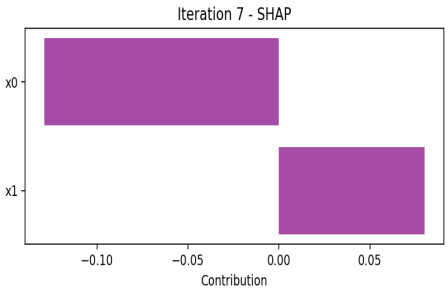
## Iteration 6 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | 0.08889 |
| x1 | -0.08460 |

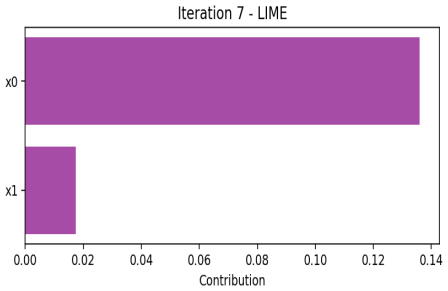
## Iteration 6 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.20042 |
| x1 | 0.04030 |

## Iteration 7 - SHAP




## Iteration 7 - LIME




## Iteration 7 - SHAP Values

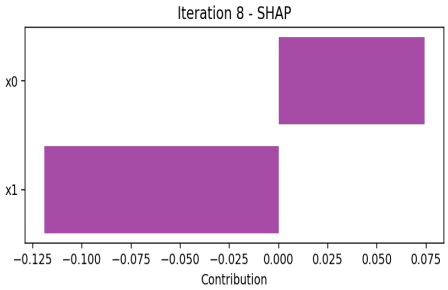
| Feature | SHAP Value |
|---|---|
| x0 | -0.12888 |
| x1 | 0.08012 |

## Iteration 7 - LIME Values

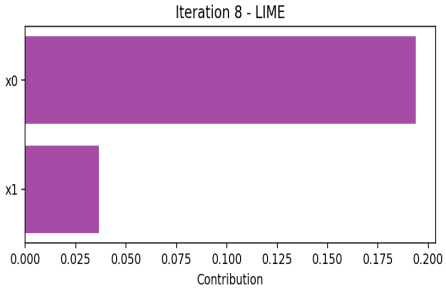
| Feature | LIME Value |
|---|---|
| x0 | 0.13596 |

| | |
|---|---|
| x1 | 0.01730 |

## Iteration 8 - SHAP




## Iteration 8 - LIME



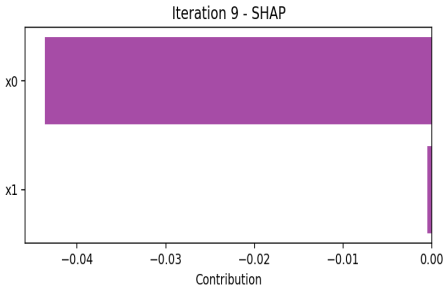

## Iteration 8 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | 0.07422 |
| x1 | -0.11907 |

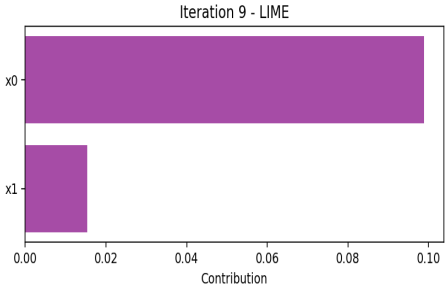
## Iteration 8 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.19383 |
| x1 | 0.03646 |

## Iteration 9 - SHAP




## Iteration 9 - LIME



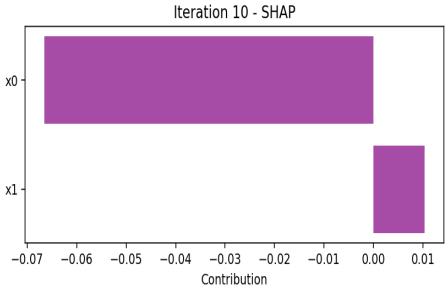

## Iteration 9 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.04364 |
| x1 | -0.00051 |

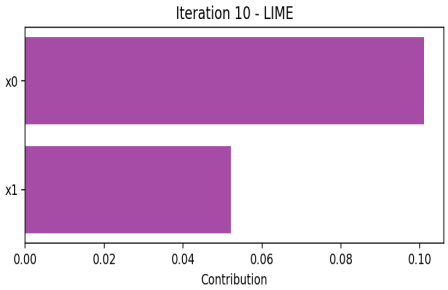
## Iteration 9 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.09885 |
| x1 | 0.01534 |

## Iteration 10 - SHAP




## Iteration 10 - LIME



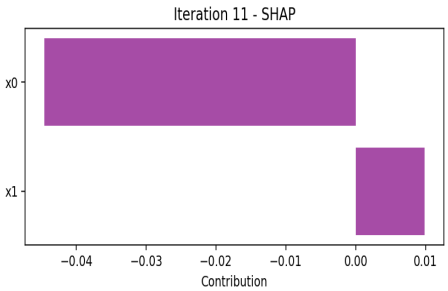

## Iteration 10 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.06656 |
| x1 | 0.01039 |

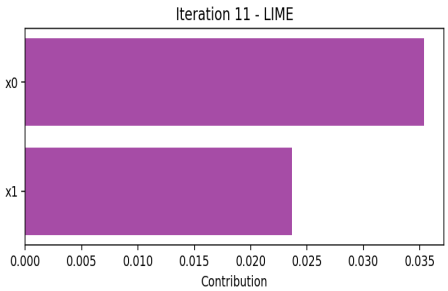
## Iteration 10 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.10105 |
| x1 | 0.05215 |

## Iteration 11 - SHAP




## Iteration 11 - LIME



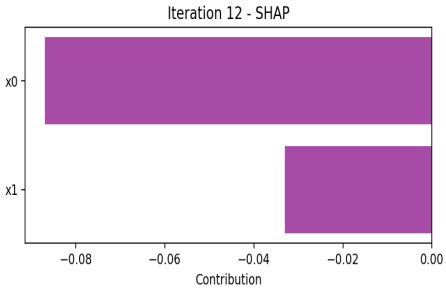

## Iteration 11 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.04460 |
| x1 | 0.00990 |

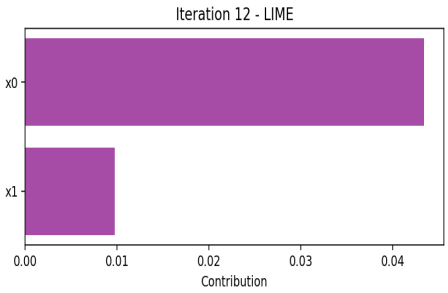
## Iteration 11 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.03537 |
| x1 | 0.02366 |

## Iteration 12 - SHAP




## Iteration 12 - LIME



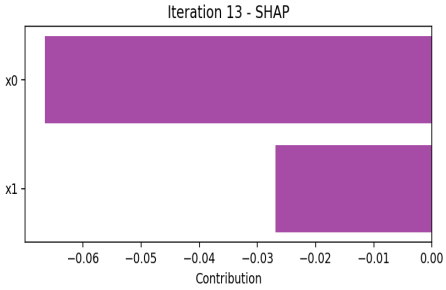

## Iteration 12 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.08699 |
| x1 | -0.03311 |

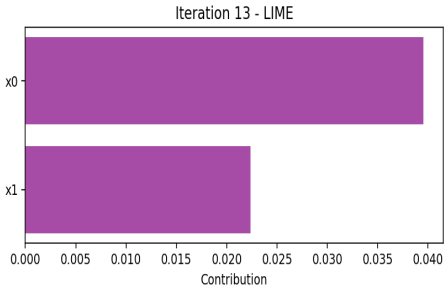
## Iteration 12 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.04339 |
| x1 | 0.00973 |

## Iteration 13 - SHAP




## Iteration 13 - LIME



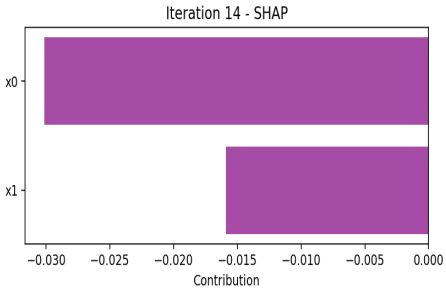

## Iteration 13 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.06659 |
| x1 | -0.02694 |

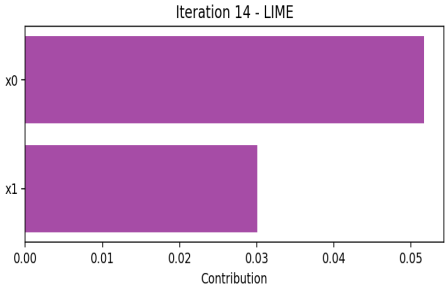
## Iteration 13 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.03954 |
| x1 | 0.02238 |

## Iteration 14 - SHAP



## Iteration 14 - LIME



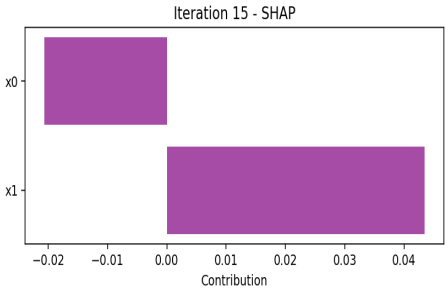

## Iteration 14 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.03013 |
| x1 | -0.01592 |

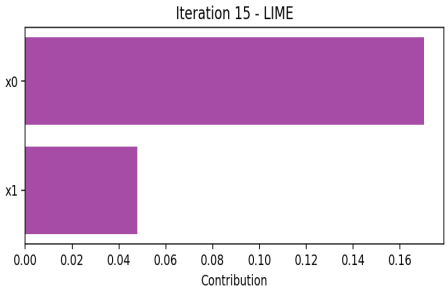
## Iteration 14 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.05165 |
| x1 | 0.03005 |

## Iteration 15 - SHAP




## Iteration 15 - LIME



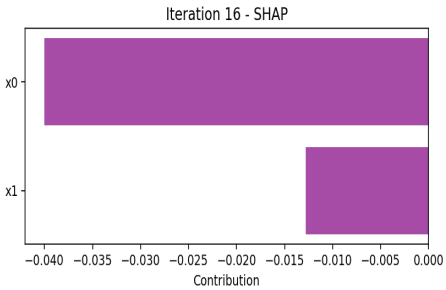
## Iteration 15 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.02067 |
| x1 | 0.04350 |

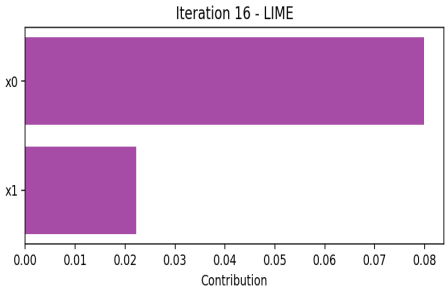
## Iteration 15 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.17024 |
| x1 | 0.04782 |

## Iteration 16 - SHAP




## Iteration 16 - LIME




## Iteration 16 - SHAP Values

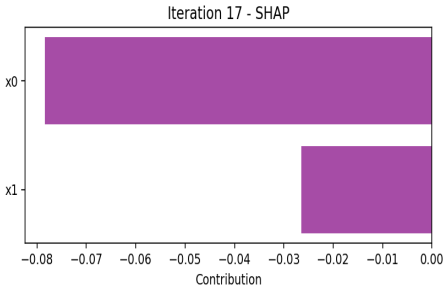
| Feature | SHAP Value |
|---|---|
| x0 | -0.04000 |
| x1 | -0.01280 |

## Iteration 16 - LIME Values

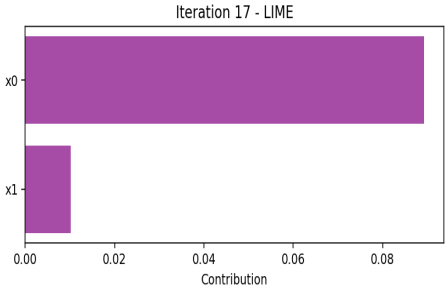
| Feature | LIME Value |
|---|---|
| x0 | 0.07986 |

| | |
|---|---|
| x1 | 0.02223 |

## Iteration 17 - SHAP




## Iteration 17 - LIME



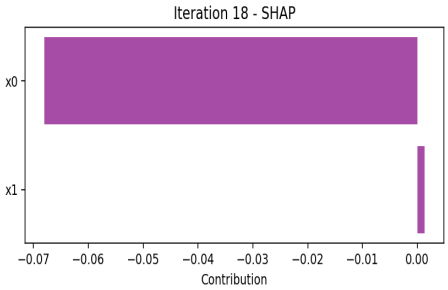

## Iteration 17 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.07843 |
| x1 | -0.02647 |

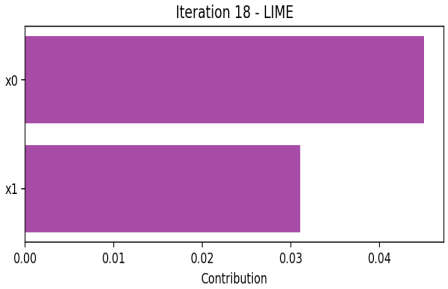
## Iteration 17 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.08925 |
| x1 | 0.01020 |

## Iteration 18 - SHAP




## Iteration 18 - LIME



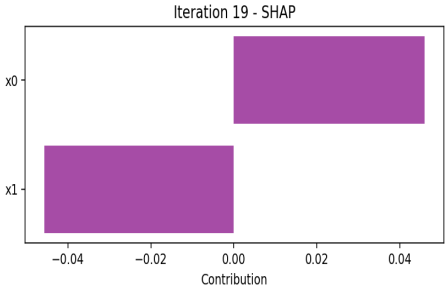

## Iteration 18 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.06804 |
| x1 | 0.00137 |

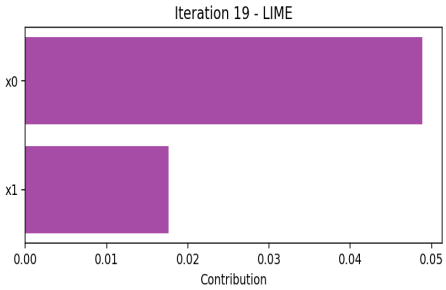
## Iteration 18 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.04500 |
| x1 | 0.03105 |

## Iteration 19 - SHAP




## Iteration 19 - LIME



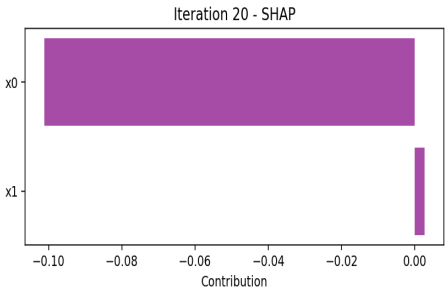

## Iteration 19 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | 0.04596 |
| x1 | -0.04564 |

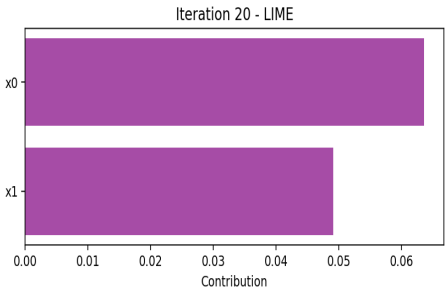
## Iteration 19 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.04889 |
| x1 | 0.01763 |

## Iteration 20 - SHAP




## Iteration 20 - LIME



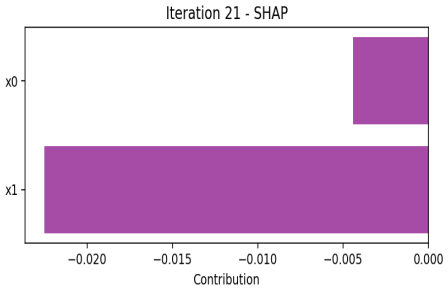

## Iteration 20 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.10121 |
| x1 | 0.00271 |

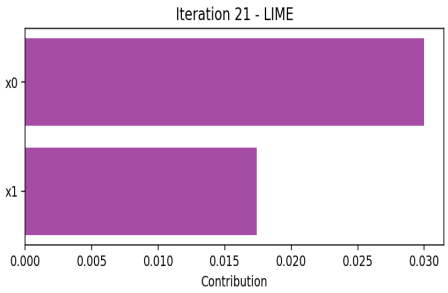
## Iteration 20 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.06358 |
| x1 | 0.04913 |

## Iteration 21 - SHAP




## Iteration 21 - LIME



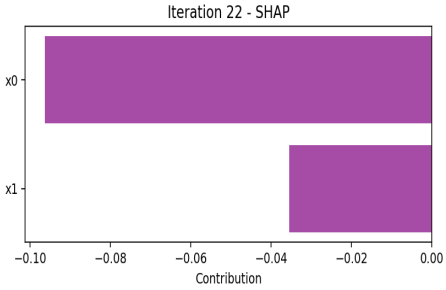

## Iteration 21 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.00442 |
| x1 | -0.02251 |

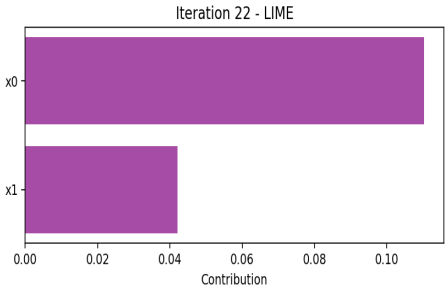
## Iteration 21 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.02997 |
| x1 | 0.01741 |

## Iteration 22 - SHAP




## Iteration 22 - LIME



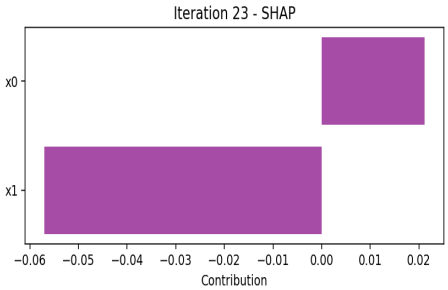

## Iteration 22 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.09640 |
| x1 | -0.03558 |

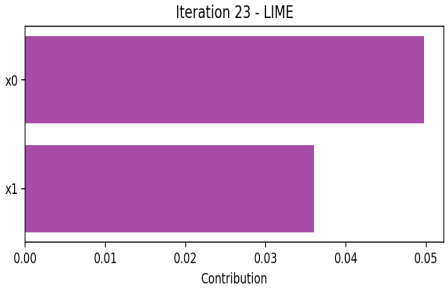
## Iteration 22 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.11024 |
| x1 | 0.04217 |

## Iteration 23 - SHAP



## Iteration 23 - LIME



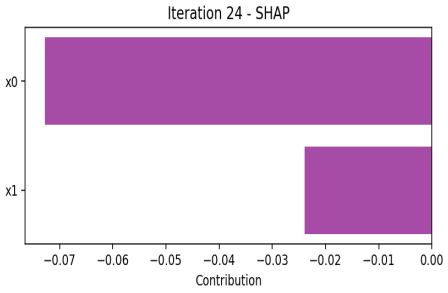

## Iteration 23 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | 0.02120 |
| x1 | -0.05702 |

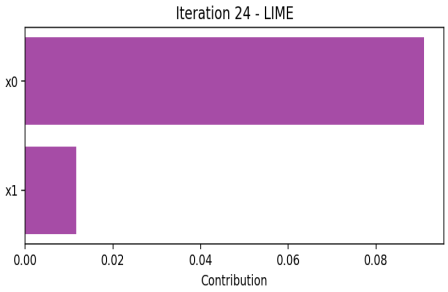
## Iteration 23 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.04973 |
| x1 | 0.03602 |

## Iteration 24 - SHAP




## Iteration 24 - LIME



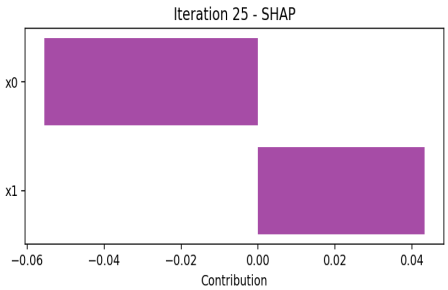
## Iteration 24 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.07280 |
| x1 | -0.02391 |

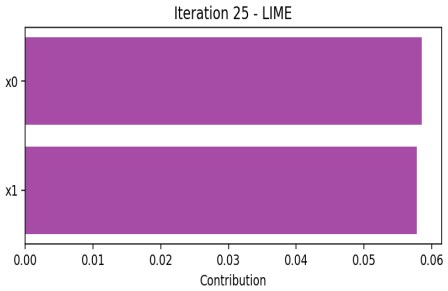
## Iteration 24 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.09092 |
| x1 | 0.01158 |

## Iteration 25 - SHAP




## Iteration 25 - LIME




## Iteration 25 - SHAP Values

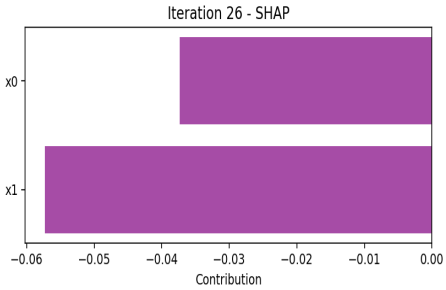
| Feature | SHAP Value |
|---|---|
| x0 | -0.05553 |
| x1 | 0.04333 |

## Iteration 25 - LIME Values

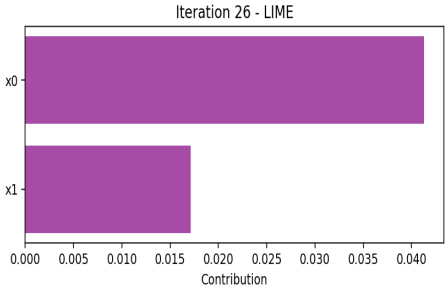
| Feature | LIME Value |
|---|---|
| x0 | 0.05850 |

| | |
|---|---|
| x1 | 0.05776 |

## Iteration 26 - SHAP




## Iteration 26 - LIME



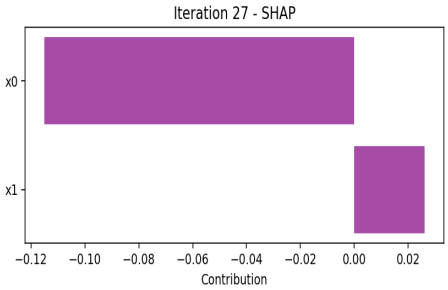

## Iteration 26 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.03736 |
| x1 | -0.05733 |

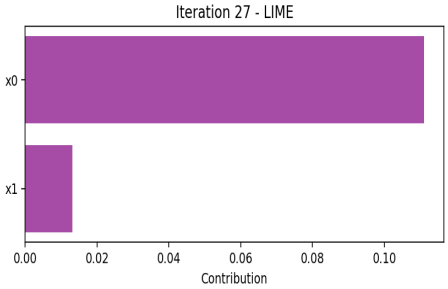
## Iteration 26 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.04128 |
| x1 | 0.01713 |

## Iteration 27 - SHAP




## Iteration 27 - LIME



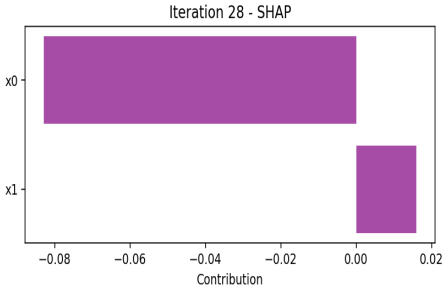

## Iteration 27 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.11509 |
| x1 | 0.02622 |

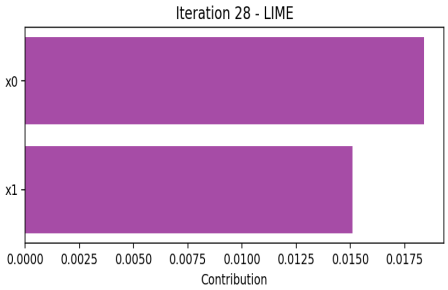
## Iteration 27 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.11102 |
| x1 | 0.01306 |

## Iteration 28 - SHAP




## Iteration 28 - LIME



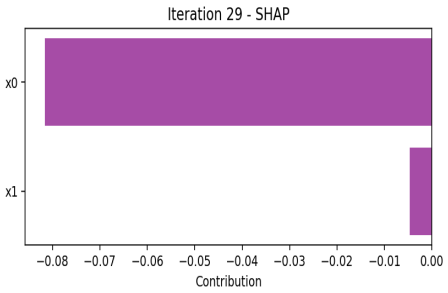

## Iteration 28 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.08282 |
| x1 | 0.01597 |

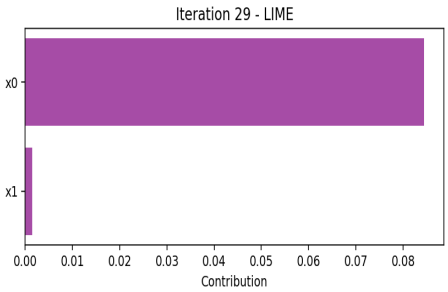
## Iteration 28 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.01838 |
| x1 | 0.01510 |

## Iteration 29 - SHAP




## Iteration 29 - LIME



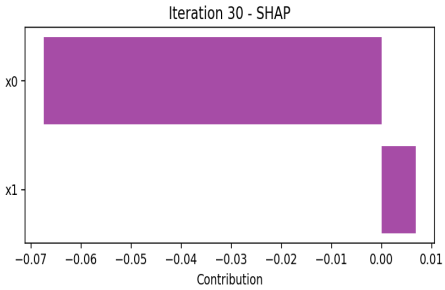

## Iteration 29 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.08164 |
| x1 | -0.00471 |

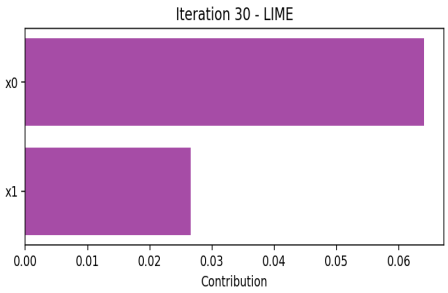
## Iteration 29 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.08441 |
| x1 | 0.00148 |

## Iteration 30 - SHAP




## Iteration 30 - LIME




## Iteration 30 - SHAP Values

| Feature | SHAP Value |
|---|---|
| x0 | -0.06740 |
| x1 | 0.00684 |

## Iteration 30 - LIME Values

| Feature | LIME Value |
|---|---|
| x0 | 0.06405 |
| x1 | 0.02658 |